MC 451 Inferencing Data in R rubric

100 points. Due Friday, November 20, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 100. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Submissions are unlimited before the deadline, so you can revise and submit again.

Inferencing Data in R, 100 points

Criterion	Exemplary	Proficient	Developing	Beginning
Hypothesis 10 points	**10 points.** Stated in plain English before any code, naming the two variables and the relationship you expect.	**8 to 9 points.** Stated before the code, but one variable is vague.	**5 to 7 points.** Stated after the results, or as a topic instead of a relationship.	**0 to 4 points.** Missing.
Test selection and its reason 20 points	**20 points.** Picks the test that fits your variable types (two categorical: chi-square; a two-group category against a continuous variable: independent t-test; two continuous: Pearson correlation) and says in one sentence why that test and not another.	**15 to 19 points.** The right test, but the reason is missing or would fit any test.	**10 to 14 points.** A test that does not fit the variable types, with a reason given.	**0 to 9 points.** No test chosen, or a test that does not fit with no reason.
Execution 15 points	**15 points.** The test runs in your script on your analysis dataset, and the output shown is what the code produces.	**11 to 14 points.** Runs, but on a different variable or subset than the hypothesis names, with no word on why.	**8 to 10 points.** The code stops with an error, or the output is pasted in instead of produced.	**0 to 7 points.** Not run.

Criterion	Exemplary	Proficient	Developing	Beginning
APA reporting 20 points	**20 points.** Reported in APA format with the test statistic, the degrees of freedom, the p-value and the n.	**15 to 19 points.** One of the four is missing, most often the n or the degrees of freedom.	**10 to 14 points.** The statistic and p only, or numbers copied from the console without APA form.	**0 to 9 points.** Not reported.
Effect size 20 points	**20 points.** Reports the effect size and says what it means in the units of your data, not only as a label such as "small".	**15 to 19 points.** The effect size and its label, with no translation into what it means for your data.	**10 to 14 points.** Computed but not explained, or the wrong measure for the test.	**0 to 9 points.** Missing: reporting p and stopping. A p-value says something happened; only the effect size says whether it matters.
Interpretation 10 points	**10 points.** Three to four sentences in plain English saying what the result means for your question.	**8 to 9 points.** Plain English, but it restates the numbers without saying what they mean for the question.	**5 to 7 points.** One or two sentences, which usually means the "so what" is missing, or a reading that treats a small p as a large effect.	**0 to 4 points.** None.
What it does not license 5 points	**5 points.** One sentence on what the result does not let you claim, such as that it holds beyond your sample.	**4 points.** A limit, but a generic one ("more research is needed").	**3 points.** A limit of the data in general, not of this result.	**0 to 2 points.** Missing.

Before you submit

Inferential Check recomputes a chi-square or a t test from your own table or group figures and returns the statistic, the degrees of freedom, p, the effect size, the assumptions your data may be breaking, and the APA sentence to compare against what you wrote. It is a second opinion on your R output, not a replacement for it. It does not grade, and nothing you enter leaves your machine.